

Lecture Notes on the Pentagonal Number Theorem

Contents

1	Introduction	2
2	Partitions and Generating Functions	2
2.1	Partitions	2
2.2	Generating function for partitions	3
3	The Pentagonal Numbers	3
3.1	Ordinary pentagonal numbers	3
3.2	Generalized pentagonal numbers	3
4	Statement of the Pentagonal Number Theorem	4
5	Why This Is Surprising	4
6	Combinatorial Meaning of the Product	4
7	A First Examples of the Coefficients	5
8	A Proof via Euler's Recurrence Argument	5
9	Euler's Recurrence for the Partition Function	6
10	Ferrers Diagrams	7
11	Durfee Squares	7
12	Euler's Involution: Outline	7
13	Description of the Exceptional Partitions	8
14	Euler's Involution in More Detail	8
15	A More Precise Involution Formulation	9
16	Proof of the Pentagonal Number Theorem	9
17	First Terms of the Expansion	10
18	Computing with the Theorem	10
19	The Theorem as an Identity of Formal Power Series	10
20	Connection with the Jacobi Triple Product	11
21	Geometric Intuition for the Pentagonal Numbers	11

22 A Table of Small Values	11
23 Applications	12
23.1 Partition recurrence	12
23.2 Congruences and modular ideas	12
23.3 Combinatorial involutions	12
24 Summary	12
25 Final Theorem Statement	13
26 Exercises	13
27 Selected Answers	13

1 Introduction

One of the most beautiful identities in the theory of partitions is Euler's *Pentagonal Number Theorem*:

$$\prod_{m=1}^{\infty} (1 - q^m) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2}.$$

At first sight this formula is mysterious. The left-hand side is an infinite product, while the right-hand side is a highly sparse power series involving the numbers

$$\frac{k(3k-1)}{2}, \quad k \in \mathbf{Z}.$$

These are the *generalized pentagonal numbers*:

$$0, 1, 2, 5, 7, 12, 15, 22, 26, \dots$$

The theorem is central in the study of partitions. It leads directly to Euler's recurrence for the partition function $p(n)$, and it is also a prototype for many deep identities in q -series, modular forms, and combinatorics.

The aim of these notes is to explain the theorem carefully, from basic definitions up to proof and applications.

2 Partitions and Generating Functions

2.1 Partitions

Definition 2.1. A *partition* of a nonnegative integer n is a way of writing n as a sum of positive integers

$$n = \lambda_1 + \lambda_2 + \dots + \lambda_r$$

with

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r \geq 1.$$

The integers λ_i are called the *parts* of the partition.

Example 2.2. The partitions of 4 are:

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

So the number of partitions of 4 is 5.

Definition 2.3. Let $p(n)$ denote the number of partitions of n , with $p(0) = 1$.

2.2 Generating function for partitions

A standard argument shows that

$$\sum_{n=0}^{\infty} p(n)q^n = \prod_{m=1}^{\infty} \frac{1}{1-q^m}.$$

Indeed, for each integer $m \geq 1$, the factor

$$\frac{1}{1-q^m} = 1 + q^m + q^{2m} + q^{3m} + \dots$$

records how many times the part m is used in a partition. Multiplying over all m gives the partition generating function.

The Pentagonal Number Theorem concerns the reciprocal product

$$\prod_{m=1}^{\infty} (1 - q^m).$$

3 The Pentagonal Numbers

3.1 Ordinary pentagonal numbers

Definition 3.1. The n -th pentagonal number is

$$P_n = \frac{n(3n-1)}{2}, \quad n \geq 1.$$

These begin as

$$1, 5, 12, 22, 35, \dots$$

They arise from a classical polygonal-number formula. For the Pentagonal Number Theorem, however, what appears naturally is a symmetrized version.

3.2 Generalized pentagonal numbers

Definition 3.2. The *generalized pentagonal numbers* are the integers

$$\frac{k(3k-1)}{2}, \quad k \in \mathbf{Z}.$$

If we list them for

$$k = 0, 1, -1, 2, -2, 3, -3, \dots$$

we get

$$0, 1, 2, 5, 7, 12, 15, 22, 26, \dots$$

Explicitly:

$$\begin{aligned} \frac{1(3 \cdot 1 - 1)}{2} &= 1, & \frac{(-1)(3(-1) - 1)}{2} &= 2, \\ \frac{2(6 - 1)}{2} &= 5, & \frac{(-2)(-6 - 1)}{2} &= 7, \end{aligned}$$

and so on.

Remark 3.3. The generalized pentagonal numbers occur in pairs:

$$\frac{k(3k-1)}{2}, \quad \frac{k(3k+1)}{2},$$

for $k \geq 1$, because

$$\frac{(-k)(3(-k) - 1)}{2} = \frac{k(3k+1)}{2}.$$

Thus the theorem is often written in the form

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(q^{k(3k-1)/2} + q^{k(3k+1)/2} \right).$$

4 Statement of the Pentagonal Number Theorem

Theorem 4.1 (Pentagonal Number Theorem). *In the ring of formal power series,*

$$\prod_{m=1}^{\infty} (1 - q^m) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2}.$$

Equivalently,

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(q^{k(3k-1)/2} + q^{k(3k+1)/2} \right).$$

Expanding the first terms gives

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + q^{22} + q^{26} - \dots$$

There are no other powers of q : only the generalized pentagonal numbers appear.

5 Why This Is Surprising

The finite partial products

$$(1 - q)(1 - q^2)(1 - q^3) \cdots (1 - q^N)$$

expand into many terms, with cancellations. It is far from obvious that in the infinite limit the remaining coefficients are so sparse and so structured.

The theorem says:

- if n is not a generalized pentagonal number, then the coefficient of q^n is 0;
- if

$$n = \frac{k(3k \pm 1)}{2},$$

then the coefficient is $(-1)^k$.

So the whole power series is completely determined by this remarkable pattern.

6 Combinatorial Meaning of the Product

Let us interpret

$$\prod_{m=1}^{\infty} (1 - q^m)$$

combinatorially.

For each m , the factor

$$1 - q^m$$

means: either do not choose the part m , or choose it once and attach a minus sign.

Therefore the coefficient of q^n in the product is

$$\sum_{\lambda} (-1)^{\ell(\lambda)},$$

where the sum runs over partitions λ of n into *distinct* parts, and $\ell(\lambda)$ denotes the number of parts.

So the coefficient of q^n is:

(number of partitions of n into an even number of distinct parts) – (number of partitions of n into an odd number of distinct parts).

The Pentagonal Number Theorem therefore says that this signed difference is almost always 0, except at the generalized pentagonal numbers.

7 A First Examples of the Coefficients

Example 7.1. For $n = 3$, the partitions into distinct parts are

$$3, \quad 2 + 1.$$

So the signed contribution is

$$(-1)^1 + (-1)^2 = -1 + 1 = 0.$$

Indeed, 3 is not a generalized pentagonal number.

Example 7.2. For $n = 5$, the partitions into distinct parts are

$$5, \quad 4 + 1, \quad 3 + 2.$$

The signed sum is

$$(-1)^1 + (-1)^2 + (-1)^2 = -1 + 1 + 1 = 1.$$

Since $5 = \frac{2(3 \cdot 2 - 1)}{2}$, the theorem predicts coefficient +1.

Example 7.3. For $n = 6$, the partitions into distinct parts are

$$6, \quad 5 + 1, \quad 4 + 2, \quad 3 + 2 + 1.$$

The signed sum is

$$-1 + 1 + 1 - 1 = 0.$$

Again this agrees with the theorem.

8 A Proof via Euler's Recurrence Argument

We first explain the theorem in a way closely tied to partitions and generating functions. Then later we give the classic involution proof.

Let

$$F(q) = \prod_{m=1}^{\infty} (1 - q^m).$$

Suppose

$$F(q) = \sum_{n=0}^{\infty} a_n q^n.$$

Then

$$\left(\sum_{n=0}^{\infty} p(n)q^n \right) \left(\sum_{n=0}^{\infty} a_n q^n \right) = 1,$$

because

$$\sum_{n=0}^{\infty} p(n)q^n = \prod_{m=1}^{\infty} \frac{1}{1 - q^m}.$$

Thus the coefficients a_n are determined by the condition that

$$\sum_{k=0}^n p(k)a_{n-k} = 0 \quad (n \geq 1),$$

with $a_0 = 1$.

Euler observed experimentally that the a_n are exactly

$$a_n = \begin{cases} (-1)^m, & n = \frac{m(3m-1)}{2} \text{ or } \frac{m(3m+1)}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

This leads to the partition recurrence

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) - \dots$$

where the offsets are generalized pentagonal numbers.

This recurrence is one of the main uses of the theorem.

9 Euler's Recurrence for the Partition Function

Corollary 9.1 (Euler recurrence). *For $n \geq 1$,*

$$p(n) = \sum_{k=1}^{\infty} (-1)^{k-1} \left(p\left(n - \frac{k(3k-1)}{2}\right) + p\left(n - \frac{k(3k+1)}{2}\right) \right),$$

where $p(m) = 0$ for $m < 0$.

Proof. Starting from

$$\left(\sum_{n=0}^{\infty} p(n)q^n \right) \left(1 + \sum_{k=1}^{\infty} (-1)^k \left(q^{k(3k-1)/2} + q^{k(3k+1)/2} \right) \right) = 1,$$

compare coefficients of q^n for $n \geq 1$. □

Example 9.2. Let us compute $p(5)$:

$$p(5) = p(4) + p(3) - p(0).$$

Since

$$p(4) = 5, \quad p(3) = 3, \quad p(0) = 1,$$

we get

$$p(5) = 5 + 3 - 1 = 7.$$

10 Ferrers Diagrams

To explain the combinatorial proof, we need diagrams.

Definition 10.1. The *Ferrers diagram* of a partition

$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_r)$$

is the array of dots or boxes with λ_i boxes in row i , left-justified.

Example 10.2. The partition

$$7 = 4 + 2 + 1$$

has Ferrers diagram

$$\begin{array}{cccc} \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & & \\ \bullet & & & \end{array}$$

For partitions into distinct parts, the row lengths are strictly decreasing.

11 Durfee Squares

Definition 11.1. The *Durfee square* of a partition is the largest square that fits inside its Ferrers diagram.

Example 11.2. For

$$\lambda = (6, 4, 3, 1),$$

the Durfee square has side length 3, since the first three rows have lengths at least 3, but a 4×4 square does not fit.

Durfee squares are useful in the involution proof because they organize partitions according to their geometric shape.

12 Euler's Involution: Outline

Let $D(n)$ be the set of partitions of n into distinct parts. We want to evaluate

$$\sum_{\lambda \in D(n)} (-1)^{\ell(\lambda)}.$$

Euler's idea is to pair off most partitions in $D(n)$ so that each pair contributes

$$(-1)^r + (-1)^{r+1} = 0.$$

Only a small exceptional family remains unmatched, and those exceptional partitions occur exactly when n is a generalized pentagonal number.

That is the source of the theorem.

13 Description of the Exceptional Partitions

The unmatched partitions in Euler's involution turn out to be of the forms

$$(2m-1, 2m-2, \dots, m)$$

and

$$(2m, 2m-1, \dots, m+1).$$

Let us compute their sizes.

For the first:

$$(2m-1) + (2m-2) + \dots + m = \sum_{j=m}^{2m-1} j.$$

This is

$$\frac{(m + (2m-1))m}{2} = \frac{(3m-1)m}{2}.$$

For the second:

$$(2m) + (2m-1) + \dots + (m+1) = \sum_{j=m+1}^{2m} j = \frac{((m+1) + 2m)m}{2} = \frac{m(3m+1)}{2}.$$

These are exactly the two forms of generalized pentagonal numbers.

14 Euler's Involution in More Detail

We now sketch the classic involution. Let

$$\lambda = (\lambda_1 > \lambda_2 > \dots > \lambda_r)$$

be a partition into distinct parts.

Define:

- $s = r$, the number of parts;
- $t = \lambda_r$, the smallest part.

The Ferrers diagram has a diagonal of some length, and one compares the smallest part with the number of parts. Roughly:

- if the smallest part is larger than the number of parts, remove 1 from each part and append a new smallest part;
- if the smallest part is at most the number of parts, reverse the process by deleting the smallest part and adding 1 to several parts.

This operation changes the number of parts by 1, hence changes the sign. It is an involution on all non-exceptional partitions.

The exceptional cases are precisely the staircase-like partitions described above, and these yield the generalized pentagonal numbers.

15 A More Precise Involution Formulation

Let

$$\lambda = (\lambda_1 > \lambda_2 > \cdots > \lambda_r)$$

be a partition into distinct parts.

There is a standard map defined on non-exceptional partitions:

1. If $\lambda_r > r$, replace

$$(\lambda_1, \lambda_2, \dots, \lambda_r)$$

by

$$(\lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_r - 1, r).$$

2. If $\lambda_r \leq r$, reverse this process.

One checks:

- the resulting partition still has distinct parts except in the exceptional staircase cases;
- applying the map twice returns to the original partition;
- the parity of the number of parts changes.

Hence all non-exceptional partitions cancel in the signed sum.

The only survivors are the two staircase families:

$$(2m - 1, 2m - 2, \dots, m), \quad (2m, 2m - 1, \dots, m + 1).$$

These have m parts, so each contributes sign $(-1)^m$.

Thus

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 + \sum_{m=1}^{\infty} (-1)^m \left(q^{m(3m-1)/2} + q^{m(3m+1)/2} \right).$$

This proves the theorem.

16 Proof of the Pentagonal Number Theorem

Combinatorial proof. Expand

$$\prod_{m=1}^{\infty} (1 - q^m)$$

as a generating function for partitions into distinct parts, with sign $(-1)^{\ell(\lambda)}$. Thus the coefficient of q^n is

$$\sum_{\lambda \in D(n)} (-1)^{\ell(\lambda)}.$$

Euler's involution pairs all non-exceptional partitions in $D(n)$ into pairs whose numbers of parts differ by 1. Therefore each pair contributes zero to the signed sum.

The only partitions not paired are the two staircase families:

$$(2m - 1, 2m - 2, \dots, m), \quad (2m, 2m - 1, \dots, m + 1).$$

These have sizes

$$\frac{m(3m-1)}{2}, \quad \frac{m(3m+1)}{2},$$

respectively, and each has m parts, hence contributes sign $(-1)^m$.

Therefore

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 + \sum_{m=1}^{\infty} (-1)^m \left(q^{m(3m-1)/2} + q^{m(3m+1)/2} \right),$$

which is exactly the Pentagonal Number Theorem. □

17 First Terms of the Expansion

Let us write out the first terms:

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + q^{22} + q^{26} - \dots$$

The exponents are:

$$0, 1, 2, 5, 7, 12, 15, 22, 26, \dots$$

coming from

$$\frac{k(3k-1)}{2}, \quad k \in \mathbf{Z}.$$

The signs occur in pairs:

$$+, -, -, +, +, -, -, +, +, \dots$$

more precisely:

$$(-1)^m \quad \text{for the pair } \frac{m(3m-1)}{2}, \frac{m(3m+1)}{2}.$$

18 Computing with the Theorem

Example 18.1. Using the theorem,

$$\prod_{m=1}^{\infty} (1 - q^m) = 1 - q - q^2 + q^5 + q^7 - \dots$$

so the coefficient of q^6 is 0.

Example 18.2. The coefficient of q^{12} is -1 , because

$$12 = \frac{3(3 \cdot 3 - 1)}{2} = \frac{3 \cdot 8}{2} = 12.$$

Example 18.3. The coefficient of q^{15} is also -1 , because

$$15 = \frac{3(3 \cdot 3 + 1)}{2}.$$

19 The Theorem as an Identity of Formal Power Series

It is important to note that the theorem is best interpreted formally. In the ring of formal power series $\mathbf{Z}[[q]]$, the infinite product

$$\prod_{m=1}^{\infty} (1 - q^m)$$

is well-defined because the coefficient of q^n depends only on finitely many factors.

So the theorem is not merely an analytic identity for $|q| < 1$, though it is true analytically there as well. It is an exact algebraic identity in formal power series.

Remark 19.1. This formal viewpoint is standard in generatingfunctionology and combinatorics.

20 Connection with the Jacobi Triple Product

The Pentagonal Number Theorem can also be derived from the Jacobi triple product:

$$\sum_{k=-\infty}^{\infty} z^k q^{k^2} = \prod_{m=1}^{\infty} (1 - q^{2m})(1 + zq^{2m-1})(1 + z^{-1}q^{2m-1}).$$

By a suitable specialization, one obtains Euler's identity. This places the Pentagonal Number Theorem in a broader family of theta-function identities.

Although we do not pursue that proof here, it is worth knowing that Euler's theorem sits at the entrance to a vast subject.

21 Geometric Intuition for the Pentagonal Numbers

Why pentagonal numbers? The short answer is: not because pentagons are visually built into the product, but because the exceptional staircase partitions have sizes

$$\frac{m(3m \pm 1)}{2},$$

which are exactly the generalized pentagonal numbers.

So the pentagonal numbers arise from the geometry of Ferrers diagrams under Euler's involution, rather than from any immediate polygonal construction on the product side.

22 A Table of Small Values

The beginning of

$$\prod_{m=1}^{\infty} (1 - q^m)$$

looks like this:

n	$[q^n] \prod_{m=1}^{\infty} (1 - q^m)$
0	1
1	-1
2	-1
3	0
4	0
5	1
6	0
7	1
8	0
9	0
10	0
11	0
12	-1
13	0
14	0
15	-1

This matches the theorem perfectly.

23 Applications

23.1 Partition recurrence

The main application is Euler's recurrence for $p(n)$, already discussed.

23.2 Congruences and modular ideas

The infinite products

$$\prod_{m=1}^{\infty} (1 - q^m) \quad \text{and} \quad \prod_{m=1}^{\infty} \frac{1}{1 - q^m}$$

are fundamental in modular form theory and in the study of partition congruences such as Ramanujan's:

$$\begin{aligned} p(5n + 4) &\equiv 0 \pmod{5}, \\ p(7n + 5) &\equiv 0 \pmod{7}, \\ p(11n + 6) &\equiv 0 \pmod{11}. \end{aligned}$$

23.3 Combinatorial involutions

The theorem is a classical example of a *sign-reversing involution*, one of the central techniques in enumerative combinatorics.

24 Summary

Let us summarize the main points.

1. The generating function for partitions is

$$\sum_{n=0}^{\infty} p(n)q^n = \prod_{m=1}^{\infty} \frac{1}{1 - q^m}.$$

2. The reciprocal product

$$\prod_{m=1}^{\infty} (1 - q^m)$$

is the signed generating function for partitions into distinct parts.

3. Euler's involution pairs off almost all such partitions, leaving only the exceptional staircase partitions.
4. The sizes of these exceptional partitions are the generalized pentagonal numbers

$$\frac{k(3k - 1)}{2}, \quad k \in \mathbf{Z}.$$

5. Hence

$$\prod_{m=1}^{\infty} (1 - q^m) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2}.$$

That is the Pentagonal Number Theorem.

25 Final Theorem Statement

Theorem 25.1 (Euler's Pentagonal Number Theorem). *In $\mathbf{Z}[[q]]$,*

$$\prod_{m=1}^{\infty} (1 - q^m) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2} = 1 + \sum_{k=1}^{\infty} (-1)^k \left(q^{k(3k-1)/2} + q^{k(3k+1)/2} \right).$$

26 Exercises

Exercise 26.1. List all partitions of 6 into distinct parts and verify that their signed count is 0.

Exercise 26.2. Verify directly that the coefficient of q^7 in

$$\prod_{m=1}^{\infty} (1 - q^m)$$

is 1.

Exercise 26.3. Show that the partition

$$(2m-1, 2m-2, \dots, m)$$

has size $m(3m-1)/2$.

Exercise 26.4. Show that the partition

$$(2m, 2m-1, \dots, m+1)$$

has size $m(3m+1)/2$.

Exercise 26.5. Use the Pentagonal Number Theorem to compute $p(6)$ from Euler's recurrence.

Exercise 26.6. Write out the first ten generalized pentagonal numbers.

Exercise 26.7. Explain why the Pentagonal Number Theorem is naturally a statement in formal power series.

27 Selected Answers

- The first generalized pentagonal numbers are

$$0, 1, 2, 5, 7, 12, 15, 22, 26, 35.$$

- For $m = 3$,

$$(5, 4, 3)$$

has size

$$5 + 4 + 3 = 12 = \frac{3(3 \cdot 3 - 1)}{2}.$$

- Also for $m = 3$,

$$(6, 5, 4)$$

has size

$$6 + 5 + 4 = 15 = \frac{3(3 \cdot 3 + 1)}{2}.$$

One-Sentence Takeaway

Euler's Pentagonal Number Theorem says that the infinite product $\prod_{m \geq 1} (1 - q^m)$, which is the signed generating function for partitions into distinct parts, collapses to a sparse series supported exactly on the generalized pentagonal numbers.